

CORNERSTONE PDK Monitor — User Manual v1.0.2

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Version: v1.0.2 **Single-file browser dashboard for silicon photonics PDK management.**
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1. What this is

CORNERSTONE PDK Monitor is a one-page dashboard that keeps every artefact in a silicon photonics process design kit visible, browsable, and maintainable from a single browser tab. It runs entirely offline as one self-contained HTML file: no server, no install, no toolchain.

A PDK is a moving target. Layout files, S-parameter sweeps, fab process documents, EDA wrappers, post-processing scripts, supplier test reports, and design notes accumulate across years and across teams. The Monitor pulls all of that into one consistent shape: each **Technology Platform** holds a list of **Building Blocks (BBs)** — grating couplers, MMI splitters, ring filters, modulators, etc. — and every BB carries its own layout, simulation data, scripts, tests, fab notes, EDA links, variants, and documents.

A second tab kind, **DOCS**, gives the same BB structure to design manuals, app notes, fab guides, and supplier documents, so a design manual chapter behaves like a first-class entity rather than a loose file attachment.

Everything you build inside the dashboard is held in a single JSON document (dashboard.json). You can export it, share it, version it, or pull a colleague's view in and merge it with your own.

2. Quick start

1. **Open** `cornerstone-pdk-monitor-v1.0.2.html` in Chrome, Firefox, Safari, or Edge. No server required.
2. **Accept** the licence terms on first launch.
3. The dashboard loads pre-populated with the full CORNERSTONE BB set (SOI 220 nm, SiN 200 nm, SiN 400 nm, etc.). Browse tabs, click any BB card to open its detail view.
4. To **sign in**, click the user icon in the header. Choose your role:
 - **Read-only** — browse and search; no edits.
 - **User-admin** — edit your own tabs and BBs, fetch from GitHub.
 - **Full Admin** — full edit access including tab merging and dataset replacement.

5. To **start from a clean slate**, click *Import* in the header and pick the bundled `dashboard.json`, then choose *Replace whole dashboard*. This restores the factory CORNERSTONE seed.

That is the whole onboarding. Everything else is discovery from the in-app UI.

3. The dashboard layout

Header

Left to right:

- **Tabs strip** — one tab per Technology Platform or DOCS bundle. Each tab shows a small TECH or DOCS pill so you can tell at a glance which kind it is. The active tab is highlighted in the accent colour. Right-click (or long-press on mobile) a tab to rename it, merge it into another tab, delete it, or change its pill.
- **Global search box** — type two or more characters to filter the current tab. A cross-tab dropdown appears beneath the box showing matches from every other tab; click any result to jump to that tab and scroll the BB into view.
- **Tool buttons** — Import, Export, Manage Lists, Distribution, the Activity Log, the Insight agent panel, and the user/sign-in menu.

Body

The body shows the active tab. A tab contains one or more **groups** (e.g. “Active devices” / “Passive devices”). Each group renders as a card holding rows of BBs. Each row shows:

- A coloured **status box** (left edge) — Available / In progress / Deprecated / etc. Configurable per dashboard.
- The BB **name** with an inline (owner) faded suffix when the BB carries an attribution.
- A **pin** button — pins this BB to the top of its group. Pinned items always sort first; ties keep their original order.
- A **selection checkbox** for bulk actions.
- A small chevron / arrow — expands the row inline or opens the full detail panel.

Footer

Version stamp, status counts, and the storage indicator (warns before you cross IndexedDB quota limits).

4. Working with Building Blocks

Click any BB row to open its detail panel. The panel is a tabbed view; the same tabs are available for both TECH BBs and DOCS BBs.

Details

Name, identifier, status, ports / pin count, description, owner, links. Everything here is editable inline if your session has write access.

Visualisation (Viz)

Renders a layout preview when an SVG / GDS preview is available. For DOCS entries, this is where a chapter's headline image lives.

Scripts

A list of Python scripts attached to the BB. Each script has its own card. Click ► to run inside the embedded Pyodide interpreter. The globals tests (parsed test data) and data (S-matrix datasets) are exposed automatically. To emit a chart, write your figure to a buffer and `print("DASH_PLOT_PNG:" + b64(...))` — the helper `emit(fig)` does it for you. Output is captured below the editor, with PNG previews rendered inline.

You can also upload binary executables / tools as scripts; the run command is generated from the filename and editable. Pin a script to keep it at the top of the list.

Data (S-matrix / Tests)

The Data tab holds S-parameter datasets, sweep files (TXT, S2P, CSV, NPZ), and parsed metadata. Each dataset shows:

- A small preview chart with selectable metric (transmission, reflection, IL, etc.).
- File metadata (uploader, date, size, source repo).
- Pin and bulk-select controls.

Click *Detach* on the chart to pop it into a floating, resizable window — useful when you want to compare two BBs side by side. The detached chart keeps its own PNG download button.

Tests

Supplier and characterisation test reports. Same upload + pin + preview pattern as the Data tab, with per-test result tables.

Variants

Process- or layout-variants of the same BB (e.g. radius sweeps, different bend types). Each variant card links back to the parent BB and shows its own data tab.

Fab

Sub-tabs per fab process step or per supplier. Drop documents, images, photos, and notes inside; reorder by pinning.

EDA

Links to external EDA tool definitions (PCells, KLayout macros, IPKISS modules, etc.). One row per tool. The EDA list now sorts pinned entries first, so the wrappers you actually use stay at the top.

AI Insight

A built-in chat panel that can read the files attached to the current BB and answer questions about them. Configure your Anthropic or OpenAI API key once via the ⚙️ Configure button; the key is held in browser storage and never round-trips to any server other than the model provider you chose.

Use the *Data files in context* expander to selectively load test CSVs / JSON files into the conversation. Detach the panel (⌵ button) to float it next to the BB you're discussing.

5. Pinning

Anywhere a list of items lives — files, folders, EDA links, Fab sub-tabs, scripts, documents, BBs — a small / pushpin lets you keep important items at the top. Both states are styled in a neutral grey so the pin never visually competes with the content. Pin state travels with the dashboard, so what you pin stays pinned when colleagues fetch your slice.

6. Search and keyboard

- / or Cmd/Ctrl + K — focus the global search box.
- t — switch to the next tab (wraps to the first).
- n — add a new BB to the first group of the active tab (requires write access).
- ? — open the keyboard-shortcut cheatsheet.
- Esc — close the active modal, dropdown, or floating window.

Shortcuts are suppressed while you're typing in any input, textarea, or contenteditable, so they never collide with normal editing.

The cross-tab search dropdown caps at 30 results to stay responsive on large dashboards. Click any result to teleport to it.

7. Importing, exporting, distribution

Export

Click *Export* in the header to download your current state as dashboard.json. The file carries everything the Monitor knows — BBs, files, parsed data, pin state, owner attribution, the activity log, and your local lists.

Import

Click *Import* and choose a JSON file. Two modes:

- **Merge** — incoming groups are joined with yours by name; new BBs append, existing BBs update in place. Pin / owner state is preserved. A diff preview shows exactly which sections will be touched before you confirm.
- **Replace whole dashboard** — wipes your current state and reloads from the import. An automatic backup of your previous state is written to your downloads first, so a bad import is reversible.

Every import path schema-validates the incoming payload before touching state. Malformed files are rejected with a clear error listing the specific shape problems.

Distribution

Distribution lets you fetch a published dashboard slice from a GitHub repo or a public URL and merge it into yours. Useful for pulling a colleague's updated BB set into your local copy without hand-merging files. Same diff preview and schema validation as local import. All replace operations are logged to the Activity Log.

Manage Lists

The header *Manage Lists* button opens a panel for adding, editing, fetching, and merging the small lookup lists that drive the dashboard: statuses, owners, tags, EDA tools, fab process steps, and so on. Each list is per-user (saved in your local dashboard) but can be fetched from anyone else's published view and merged with yours.

8. GitHub sync

Each tab can be wired to a GitHub repository. From the tab menu:

1. Choose *GitHub: configure*.
2. Paste the owner/repo, branch, and an optional sub-path.
3. Provide a fine-grained Personal Access Token with **Contents: Read+Write** on the specific repo you're pointing at. Classic broad-scope repo tokens are accepted but flagged with a warning — fine-grained tokens are the recommended pattern.

Once configured:

- **Fetch tab** — re-reads the repo and merges new / changed files into the matching BBs. File dedupe is by filename, so re-fetching never produces ghost duplicates: an existing file is updated in place, preserving its id, pin state, and any uploader attribution.
- **Sync BB** — same operation, scoped to one BB's files only.
- **Fetch single section** — useful when you only want, e.g., freshly updated S-matrix data without touching scripts.

Files can also be **uploaded directly** from the BB panel (drag-and-drop or file picker). Direct uploads dedupe by filename too: re-uploading replaces the bytes in place rather than producing a second copy. The same applies to scripts and auto-generated YAML descriptors.

9. Access control

The dashboard runs entirely client-side, so “access control” here means workflow gates, not server enforcement. There are three roles:

- **Read-only** — default. Can browse every tab, run searches, open detail panels, view scripts and data, manage their own local tabs (renaming, hiding) but cannot mutate shared content.
- **User-admin** — sign in with an identity you choose. Can edit BBs and files they own, fetch from GitHub, manage their own lists, and merge in published slices.
- **Full Admin** — the dashboard owner. Can tab-merge, replace the whole dashboard, prune owner identities, and run destructive operations.

The sign-in modal shows an 8-character **session trust token**. The token is generated once per browser session and stored in `sessionStorage`. Re-opening the sign-in modal later in the same session should always show the same token. If a sign-in prompt ever shows a different token, treat it as suspicious — it may be a fake prompt mounted by an injected script.

Login is rate-limited at five attempts per identity per hour.

10. Undo, Activity Log, and auditability

30-second Undo

Every destructive action — deleting a tab, a BB, a file, a manage- list row — produces a toast with an **Undo** button that stays on screen for 30 seconds. Clicking it restores the exact deleted item back into its original position with its original id, so any cross-references survive.

Activity Log

The `button` in the header opens a circular buffer of the last 100 mutations: who did what, when, on which item. The log lives inside `dashboard.json`, so when you share your state, the audit trail travels with it. Reviewing the log is the fastest way to answer “who changed this BB last week?”.

Auto-backup

Before any *Replace whole dashboard* runs, the previous state is written to your downloads folder as `dashboard.backup-<timestamp>.json`. You can always roll back by importing that file.

11. Statistics and analysis

The header chart and the *Statistics* expander aggregate counts across every tab marked **TECH**. DOCS-tabbed bundles are excluded from BB stats so that design-manual chapters don't inflate the "available BB" count. Pie segments include every TECH platform; a BB present in multiple TECH tabs is counted once per tab.

The benchmark chart on each BB lets you flip between metrics (transmission, IL, return loss, etc.) without re-uploading data. A *Download CSV / XLSX* button at the bottom of the chart writes every metric (one sheet per metric) into a single spreadsheet.

12. Mobile / tablet

The layout has two responsive breakpoints:

- **≤ 900 px** (tablet portrait, iPad on the fab floor): the header wraps onto two rows, the search box drops to its own full-width row, group cards stack one per column, modal width grows to fit the viewport.
- **≤ 600 px** (phone): tabs scroll horizontally rather than wrapping, the modal goes near-fullscreen, the activity button collapses to just its emoji, and the stats / H1 hide to free up real estate.

All controls remain usable; touch targets are sized for fingers.

13. Troubleshooting

Storage quota exceeded. Large binary uploads can saturate the IndexedDB quota the browser grants origin-less file://. Move the dashboard to a localhost server (any quick static server will do) to unlock a larger quota. The footer storage indicator warns you before you hit the wall.

Chart shows "No numeric columns." The parser couldn't find two numeric columns in the file. Open the file inline and confirm the header / delimiter is what the parser expects; the dataset editor lets you re-pick columns by index.

A fetched section seems empty after merging. If your local group's name is a prefix of an incoming empty group, the merger keeps the longer local name to avoid creating a duplicate empty group. Re-fetching the full slice repopulates it.

Sign-in keeps failing. Five attempts per identity per hour is the rate limit. Wait, or use a different identity string. The limit is per-browser, not per-IP.

Lost something I deleted, no toast in sight. Reload the page and check the Activity Log — the delete is recorded there with a timestamp; pair it with the latest dashboard.backup-*.json download to recover the entity.

14. Where things live on disk

- cornerstone-pdk-monitor-v1.0.2.html — the only file you need to open. Contains the seed dashboard, all of the app code, and every dependency.
- dashboard.json — full state restore-point. Import this to reset to factory CORNERSTONE defaults.
- USER_MANUAL.md / .docx / .pdf — this document.
- CHANGELOG.md — release notes.
- CORNERSTONE_PDK_Monitor_Introduction.pptx — slide deck for team intros.
- README.txt — short release readme.

The Monitor never writes to disk silently. Every save is an explicit export.

15. AI Insight quick start

1. Click **Insight** in the header.
 2. Click  **Configure** and paste your Anthropic or OpenAI API key. The key is held in browser storage; nothing leaves the tab except the requests you initiate.
 3. Click **Chat with the agent** to open a thread on the current BB or DOCS entry.
 4. Expand ► **Data files in context** to attach any of the BB's CSV / JSON / S-matrix files. The agent can run quick analyses on them via *Analyse files with AI*.
 5. Click **Detach** to pop the chat into a floating window so it stays visible while you navigate the rest of the dashboard.
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16. Licensing and credits

CORNERSTONE PDK Monitor is licenced under the **TAPR Open Hardware Licence v1.0**. You may redistribute, modify, and use it for any purpose; derivative works should keep the same licence.

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Project home: <https://pdk.cornerstone.soton.ac.uk> Open platform repo:
<https://github.com/cornerstone-uos/cornerstone-community> Main PDK repo:
<https://github.com/cornerstone-uos/cornerstone-pdk> Website:
<https://cornerstone.sotonfab.co.uk>

Bug reports, feature requests, and pull requests are welcome at the GitHub repos above.